

Digital Logic Design

BRANCH: ELECTRONICS (ECE)

COURSE CODE: EC202

Subject Overview & Scope

This comprehensive document serves as the official compiled study notes for Digital Logic Design (EC202) under the Electronics (ECE) department. It is structured into four core university syllabus units covering primary concepts, definitions, formulas, and advanced technical applications for university examinations and placements.

University Course Syllabus & Core Units

Unit 1: Number Systems & Boolean Algebra

Binary, Octal, Decimal, Hexadecimal representation. Code conversions, 1's and 2's complements. Boolean theorems, K-Maps (up to 5 variables), Quine-McCluskey method.

Unit 2: Combinational Logic

Design of Half Adder, Full Adder, Subtractor. Multiplexers, Demultiplexers, Decoders, Encoders. Magnitude Comparators. Code converters.

Unit 3: Sequential Circuits

Latches vs Flip-flops (SR, JK, D, T). Triggering of flip-flops. Excitation tables. Design of Counters (Asynchronous, Synchronous), Shift Registers, Ring Counter, Johnson Counter.

Unit 4: Finite State Machines & PLDs

State reduction and assignment. Mealy and Moore models. ROM, PLA, PAL implementations. Intro to FPGA.

*De Morgan's Laws: $(A + B)' = A' * B'$; $(A * B)' = A' + B'$. JK Flip-Flop characteristic: $Q(t+1) = J*Q' + K'*Q$.*

CRITICAL FORMULA / PRINCIPLE:

Key Formulas & Exam Pointers



